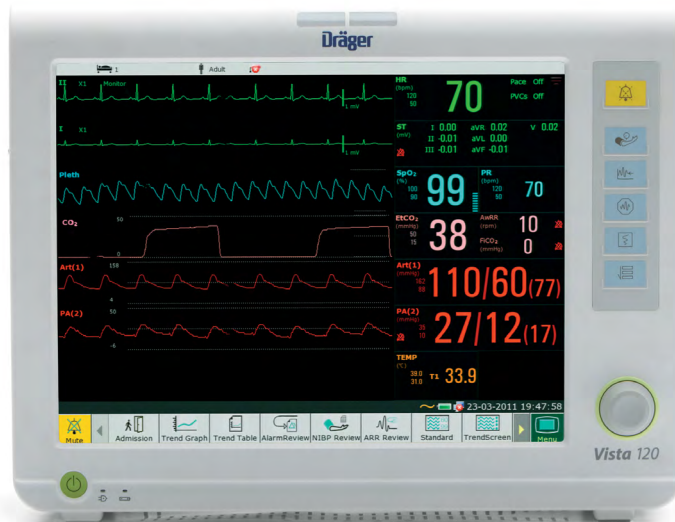




Vista 120

Patient Monitoring Solution

Hospitals around the world share a common challenge – to provide the best possible care in locations with growing populations, stricter financial regulations and caregivers that are increasingly overloaded. The Vista 120 was engineered to meet your clinical needs and stay within your budget, allowing you to deliver efficient and high-quality patient care.

Vista 120

380 mm (15") TFT touch screen

High resolution display (1,024 x 768) is bright and easy to read, even from a distance

Configurable layout

Lets you see the information you want, the way you want to see it

Enhanced trending

Stores up to 150 hours of trend data for all parameters in tabular and graphic formats Stores up to 1,200 NIBP measurements and 200 alarm events 96 hours of full disclosure

Networking capabilities

Enables central monitoring

Core set of essential parameters

3/5 lead ECG, SpO₂, non-invasive blood pressure, respiration and dual temperature

Anesthesia support

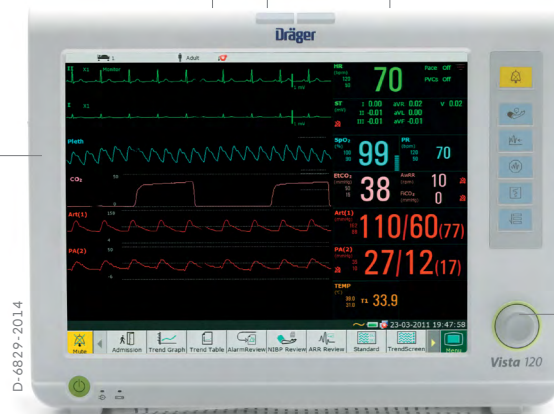
Displays data from Scio Four gas measurement modules

Alarms

Alarm indicator and alarm pause/off

Shortcut keys

Fast access to main functions



Benefits

Fully-integrated workstation solution

The Vista 120 supports adult, pediatric and neonatal patients in a variety of care environments – including Intensive Care, Operating Rooms, Emergency Departments and Neonatal Intensive Care. Medibus/Medibus-X connectivity enables the Vista 120 to be used with a complementary Dräger device, such as a ventilator or anesthesia machine allowing true integrated workstation functionality.

Essential monitoring capabilities, exceptional value

The Vista 120 displays up to 13 waveforms in an easy-to-configure layout and offers a core set of essential parameters including 3/5 lead ECG, non-invasive blood pressure, respiration and dual temperature. Advanced parameters including three invasive blood pressures, flexible mainstream and sidestream etCO₂ and cardiac output are also available. Users can add external parameter modules including SCIO, CO₂ and BIS on model C and model C+ after initial device purchase.

Supports workflow efficiency

The Vista 120 is easy to learn and easy to use. You can configure the display to see the information you want to see, the way you want to see it. Fast access keys and simplified menus put the data you need right at your fingertips.

Monitor level of consciousness with flexible Bispectral Index (BIS) measurement

The Vista 120 offers BISx measurement to support clinicians with enhanced information as they monitor the depth of anesthesia. It allows the ability to better assess patient status and quickly respond to a changing condition.

Standard built-in gas interface

The Vista 120 provides seamless connectivity to Dräger Scio anesthetic gas measurement modules delivering precise inspiratory and expiratory values.

Health level-7 (HL7) international interface

The Vista 120 offers direct connection to the hospital information system (HIS) and/or an electronic medical record in HL7 protocol or a secure connection via the Vista 120 Gateway. The ability for easy access to both of these important information files help improve workflow efficiency and reduce human error.

Dräger heritage of quality

Every life is unique. Protecting, supporting and saving lives is the foundation of our company philosophy. Our goal is to provide product and solutions that support acute care, help improve patient outcomes, reduce costs and achieve greater overall patient satisfaction.

Related Products



D-45196-2021

Vista 120 Central Monitoring System

The easy-to-use Vista 120 Central Monitoring System (CMS) lets you centrally monitor the vital signs of up to 64 patients connected to Vista 120, Vista 120 S, and Vista 120 SC monitors. This central surveillance streamlines workflows for clinicians while enhancing patient care.



D-13374-2016

Vista 120 S

Dräger understands the growing need for a patient monitor with built-in connectivity that provides essential monitoring at a good value. The Vista 120 S supports adult, pediatric and neonatal patients and can be used on its own or with a Dräger therapy device as a fully integrated workstation.



D-863-2021

Vista 120 SC

With increasing demands on clinicians, it's essential to have an easy-to-use vital signs monitor that can enhance your clinical processes and help you make informed decisions that can positively impact patient care. Providing both spot check and continuous bedside monitoring capabilities, it's the ideal monitor for your various clinical needs.

Technical Data

Classification

Protection class	Class I equipment and internal powered equipment
Degree of protection against electric shock	CF: ECG (RESP), TEMP, IBP, C.O. BF: SpO ₂ , NIBP, CO ₂ , AG, BIS
Defibrillation protection	Yes
Liquid ingress protection	IPX 1
Disinfection/sterilization method	Refer to chapter "Care and Cleaning" for details.
Mode of operation	Continuous
Compliant with standards	IEC 60601-1: 2005+A1:2012; IEC 60601-1-2: 2014; EN 60601-1: 2006+A1:2013; EN60601-1-2: 2015; IEC 60601-2-49: 2011

Supported Parameters

ECG

Lead mode	3-lead wire: I, II, III 5-lead wire: I, II, III, aVR, aVL, aVF, V
Waveform	3-lead wire: 1-channel waveform 5-lead wire: 2-channel waveform, max. seven waveforms
Lead naming style	AHA, IEC
Display sensitivity	1.25 mm/mV (x0.125), 2.5 mm/mV (x0.25), 5 mm/mV (x0.5), 10 mm/mV (x1), 20 mm/mV (x2), 40 mm/mV (x4), AUTO gain
Sweep	6.25, 12.5, 25, 50 mm/s
Bandwidth (-3dB)	Diagnosis: 0.05 to 150 Hz Monitor: 0.5 to 40 Hz Surgery: 1 to 20 Hz
CMRR (Common mode rejection ratio)	Diagnostic: > 95 dB Monitor: > 105 dB Surgery: > 105 dB
Notch	In diagnosis, monitor and surgery modes: 50 Hz/60 Hz (Notch filter can be turned on or off manually)
Differential input impedance	> 5 MΩ
Input signal range	±10 mV PP
Electrode offset potential tolerance	±800 mV
Auxiliary current (Leads off detection)	Active electrode: < 100 nA Reference electrode: < 900 nA
Recovery time after defibrillation	< 5 s (measured without electrodes as IEC60601-2-27:2011, Sect. 201.8.5.5.1 requires)
Leakage current of patient	< 10 μA
Scale signal	1 mV PP, accuracy is ±5
System noise	< 30 μV PP

Technical Data

ESU protection	Cut mode: 300 W Coagulation mode: 100 W Recovery time: ≤ 10 s
Electrosurgical interference suppression	Tested according to ANSI/AAMI EC13-2002: Sect. 5.2.9.14, Complied with ANSI/AAMI EC13:2002, Sect.4.2.9.14
Minimum input slew rate (lead II)	> 2.5 V/s
Baseline reset time	< 3 s

Pace Pulse

Pulse indicator	Pulse is marked if the requirements of IEC 60601-2-27: 2011, Sect. 201.12.1.101.12 are met: Amplitude: ± 2 mV to ± 700 mV Width: 0.1 ms to 2.0 ms Ascending time: 10 μ s to 100 μ s
Pulse rejection	Pulse is rejected if the requirements of IEC 60601-2-27: 2011, Sect. 201.12.1.101.13 are met: Amplitude: ± 2 mV to ± 700 mV Width: 0.1 ms to 2.0 ms Ascending time: 10 μ s to 100 μ s

Heart Rate

Range	ADU: 15 to 300 bpm PED/NEO: 15 to 350 bpm
Accuracy	$\pm 1\%$ or ± 1 bpm, whichever is greater
Resolution	1 bpm
Sensibility	≥ 300 μ V PP

PVC

Range	ADU: 0 to 300 PVCs/min PED/NEO: 0 to 350 PVCs/min
Resolution	1 PVCs/min

ST Value

Range	-2.0 to +2.0 mV
Accuracy	-0.8 mV to +0.8 mV: ± 0.02 mV or 10%, whichever is greater
Resolution	0.01 mV

Technical Data

HR Averaging Method

Method 1	Heart rate is computed by excluding the minimum and maximum values from the 12 most recent RR intervals and averaging the residual 10 RR intervals
Method 2	If each of three consecutive RR intervals is greater than 1,200 ms, then the four most recent RR intervals are averaged to compute the HR

Range of Sinus and SV Rhythm

Tachycardia	Adult: RR interval for 5 consecutive QRS complex ≤ 0.5 s. Pediatric/neonatal: RR interval for 5 consecutive QRS complex ≤ 0.375 s
Normal	Adult: $0.5 \text{ s} < \text{RR interval for 5 consecutive QRS complex} < 1.5 \text{ s}$. Pediatric/neonatal: $0.375 \text{ s} < \text{RR interval for 5 consecutive QRS complex} < 1 \text{ s}$
Bradycardia	Adult: RR interval for 5 consecutive QRS complex ≥ 1.5 s. Pediatric/neonatal: RR interval for 5 consecutive QRS complex ≥ 1 s

Range of Ventricular Rhythm

Ventricular tachycardia	The interval of 5 consecutive ventricular complexes is less than 600 ms
Ventricular rhythm	The interval of 5 consecutive ventricular complexes ranges from 600 ms to 1,000 ms
Ventricular bradycardia	The interval of 5 consecutive ventricular complexes is higher than 1,000 ms

Startup Time for Tachycardia

Ventricular tachycardia 1 mV 206 bpm	Gain 0.5: 10 s Gain 1.0: 10 s Gain 2.0: 10 s
Ventricular tachycardia 2 mV 195 bpm	Gain 0.5: 10 s Gain 1.0: 10 s Gain 2.0: 10 s
Response time of heart rate meter to change in HR	HR range: 80 to 120 bpm Range: within 11 s HR range: 80 to 40 bpm Range: within 11 s
Tall T-wave rejection	Complied with IEC 60601-2-27: 2011, Sect. 201.12.1.101.17 minimum recommended 1.2 mV T-wave amplitude

Technical Data

Accuracy of heart rate meter and response to irregular rhythm	Complied with IEC 60601-2-27: 2011, Sect. 201.7.9.2.9.101 b) 4). The HR value after 20 s: Ventricular bigeminy: 80 ±1 bpm Slow alternating ventricular bigeminy: 60 ±1 bpm Rapid alternating ventricular bigeminy: 120 ±1 bpm Bidirectional systoles: 91 ±1 bpm		
Time to alarm for heart rate alarm conditions	Asystole alarm: ≤ 10 s HR low alarm: ≤ 10 s HR high alarm: ≤ 10 s		
Arrhythmia analyses	Asystole	V-fib/V-tach	Couplet
	Run PVCs	PVC bigeminy	PVC trigeminy
	Vent rhythm	R on T	PVCs high
	Tachy	Brady	Missed beat
	Irr rhythm	Vent brady	Pacer not capture
	Pacer not pacing		

Respiration

Method	Impedance between RA-LL, RA-LA
Baseline impedance range	200 Ω to 2,500 Ω (with ECG cables of 1 KΩ resistance)
Measuring sensitivity	Within the baseline impedance range: 0.3 Ω
Waveform bandwidth	0.2 to 2.5 Hz (-3 dB)
RR measuring and alarm range:	Adult: 0 to 120 rpm Neo/Ped: 0 to 150 rpm
Resolution	1 rpm
Accuracy	Adult: 6 rpm to 120 rpm: ±2 rpm 0 rpm to 5 rpm: not specified Neo/Ped: 6 rpm to 150 rpm: ±2 rpm 0 rpm to 5 rpm: not specified
Gain selection	x0.25, x0.5, x1, x2, x3, x4, x5
Sweep	6.25 mm/s, 12.5 mm/s, 25 mm/s, 50 mm/s
Apnea alarm time setup	10 s, 15 s, 20 s, 25 s, 30 s, 35 s, 40 s; default value is 20 s

NIBP

Method	Oscillometric
Mode	Manual, Auto, Continuous, Sequence
Measuring interval in auto mode (unit: minutes)	1/2/2.5/3/4/5/10/15/30/60/90/120/180/240/360/480
Continuous	5 min, interval is 5 s
Measuring type	Systolic pressure, diastolic pressure, mean pressure, Pause Rate
Alarm type	SYS, DIA, MAP, PR

Technical Data

Measuring and Alarm Range

Adult mode	SYS: 40 to 270 mmHg DIA: 10 to 215 mmHg MAP: 20 to 235 mmHg
Pediatric mode	SYS: 40 to 230 mmHg DIA: 10 to 180 mmHg MAP: 20 to 195 mmHg
Neonatal mode	SYS: 40 to 135 mmHg DIA: 10 to 100 mmHg MAP: 20 to 110 mmHg
Cuff pressure measuring range	0 to 300 mmHg
Pressure resolution	1 mmHg
Maximum mean error	±5 mmHg
Maximum standard deviation	8 mmHg

Maximum Measuring Period

Adult/pediatric	120 s
Neonate	90 s
Typical measuring period	20 to 35 s (depend on HR/motion disturbance)

Overpressure Protection

Adult	297 ±3 mmHg
Pediatric	245 ±3 mmHg
Neonatal	147 ±3 mmHg

Pulse Rate

Measuring range	40 to 240 bpm
Accuracy	±3 bpm or 3.5%, whichever is larger

SpO₂

Measuring range	0 to 100%
Resolution	1%

Accuracy

Adult (including pediatric)	±2% (70 to 100% SpO ₂) Undefined (0 to 69% SpO ₂)
Neonate	±3% (70 to 100% SpO ₂) Undefined (0 to 69% SpO ₂)

Technical Data

Perfusion Index

Measuring range	0 – 10, invalid PI value is 0
Resolution	1

Pulse Rate

Measuring range	25 to 300 bpm
Resolution	1 bpm
Adjustable range of alarm limits	30 to 300 bpm
Accuracy	± 2 bpm

Nellcor Module

Measuring range	1% to 100%
Alarm range	20% to 100%
Resolution	1%
Data update period	1 s
Accuracy (70% to 100% SpO ₂):	
DS-100A,	± 3%
OXI-A/N (adult) OXI-A/N (neonate)	± 4%
D-YS (infant to adult)	± 3%
D-YS (neonate) D-YS with	± 4%
D-YSE ear clip	± 3.5%
MAX-FAST	± 2%

Pulse Rate

Measuring range	20 to 300 bpm
Resolution	1 bpm
Accuracy	± 3 bpm (20 to 250 bpm)
Sensor wavelength	Approximately 660 and 900nm
Emitted light energy	<15 mW

NOTE

Information about the wavelength range can be especially useful to clinicians (for instance, when photodynamic therapy is performed).

Temperature

Channels	2
Measuring and alarm range	0 °C to +50 °C (+32 °F to +122 °F)

Technical Data

Sensor type	YSI 2.252K/YSI 10K
Resolution	+0.1 °C (+0.1 °F)
Accuracy (without sensor)	±0.1 °C (±32.18 °F)
Refresh time	Every 1 to 2 s

IBP

Accuracy (not including sensor)	±2% or ±1 mmHg, whichever is greater
Resolution	1 mmHg

Pressure Sensor

Sensitivity	5 (µV/V/mmHg)
Impedance range	300 Ω to 3,000 Ω
Filter	DC~ 12.5 Hz; DC~ 40 Hz
Zero	Range: ±200 mmHg

Measuring and Alarm Range

Art	0 to +300 mmHg
PA	-6 to +120 mmHg
CVP/RAP/LAP/ICP	-10 to +40 mmHg
P1/P2	-50 to +300 mmHg

CO₂

Complies with ISO 80601-2-55: 2011.		
Intended patient	Adult, pediatric, neonatal	
Measure parameters	etCO ₂ , FiCO ₂ , AwRR	
Unit	mmHg, %, kPa	
Measuring range	CO ₂	0 mmHg to 150 mmHg (0% to 20%)
	AwRR	2 rpm to 150 rpm
Resolution	etCO ₂	1 mmHg
	FiCO ₂	1 mmHg
	AwRR	1 rpm

Technical Data

Accuracy	etCO ₂	±2 mmHg, 0 mmHg to 40 mmHg	Respiratory rate ≤ 60 rpm	Typical conditions: Ambient temperature: (25±3)°C Barometric pressure: (760±10) mmHg Balance gas: N ₂ Sample gas flow rate: 100 ml/min
		±5% of reading, 41 mmHg to 70 mmHg		
		±8% of reading, 71 mmHg to 100 mmHg		
		±10% of reading, 101 mmHg to 150 mmHg		
		±12% of reading or ±4 mmHg, whichever is greater	Respiratory rate > 60 rpm	All conditions
	AwRR	±1 rpm		
Drift of measure accuracy	Meets the requirements of the measure accuracy			
Sample gas flow rate	50 ml/min, 70 ml/min or 100 ml/min(default), accuracy: ±15 ml/min			
Warm-up time	Display reading within 20 s; reach to the designed accuracy within 2 minutes.			
Rise time	< 400 ms (with 2 m gas sampling tube, sample gas flowrate: 100 ml/min) < 500 ms (with 2 m gas sampling tube, sample gas flowrate: 70 ml/min) < 1000 ms (with 2 m gas sampling tube, sample gas flowrate: 50 ml/min)			
Response time	< 4 s (water trap with 2 m gas sampling tube, sample gas flowrate: 100 ml/min) < 4 s (with 2 m gas sampling tube, sample gas flowrate: 70 ml/min) < 5.5 s (with 2 m gas sampling tube, sample gas flowrate: 50 ml/min)			
Work mode	Standby, measure			
O ₂ compensation	Range: 0% to 100% Resolution: 1% Default: 16%			
N ₂ O compensation	Range: 0% to 100% Resolution: 1% Default: 0%			
AG compensation	Range: 0% to 20% Resolution: 0.1% Default: 0%			
Humidity compensation method	ATPD(default), BTPS			
Barometric pressure compensation	Automatic (The change of barometric pressure will not add additional errors to the measurement values.)			
Zero calibration	Support			
Calibration	Support			
Alarm	etCO ₂ , FiCO ₂ , AwRR			
Apnea alarm delay	10 s, 15 s, 20 s, 25 s, 30 s, 35 s, 40 s, 60 s; default value is 20 s.			
Data sample rate	100 Hz			

Technical Data

etCO ₂ change ¹	AwRR ≤ 20 rpm, meet the accuracy mentioned above; AwRR > 20 rpm, etCO ₂ descends 8%; AwRR > 40 rpm, not applicable; AwRR ≤ 80 rpm, meet the accuracy mentioned above; AwRR > 80 rpm, etCO ₂ descending 8% AwRR > 120 rpm, etCO ₂ descending 10% AwRR ≤ 60 rpm, meet the accuracy mentioned above; AwRR > 60 rpm, EtCO ₂ descends 8%; AwRR > 90 rpm, EtCO ₂ descends 10%; AwRR > 120 rpm, EtCO ₂ descends 15%;
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NOTE

Use a test device equivalent to EN ISO 80601-2-55 fig 201.101 to measure at 1:2 I/E ratio. Respiration rate accuracy is determined by frequency of device, and ET READING change refers to the nominal value.

Interfering Gas Effects:

Gas	Gas Level (%)	Quantitative Effect/Comments
Nitrous oxide	60	The interfering gas will have no effect on the measurement value if compensation of O ₂ , N ₂ O, anesthetic agents has been correctly set.
Halothane	4	
Enflurane	5	
Isoflurane	5	
Sevoflurane	5	
Desflurane	15	

Respironics Module

Applicable patient type	Adult, pediatric and neonatal patients
Technique	Infra-red absorption technique
Measure parameters	etCO ₂ , FiCO ₂ , AwRR
Unit	mmHg, %, Kpa

Measuring Range

etCO ₂	0 mmHg to 150 mmHg	
FiCO ₂	3 mmHg to 50 mmHg	
AwRR	0 rpm to 150 rpm (mainstream) 2 rpm to 150 rpm (sidestream)	
Resolution	etCO ₂	1 mmHg
	FiCO ₂	1 mmHg
	AwRR	1 rpm

Technical Data

etCO ₂ accuracy	± 2 mmHg, 0 mmHg to 40 mmHg $\pm 5\%$ of reading, 41 mmHg to 70 mmHg $\pm 8\%$ of reading, 71 mmHg to 100 mmHg $\pm 10\%$ of reading, 101 mmHg to 150 mmHg $\pm 12\%$ of reading, RR is over 80 rpm (sidestream) There will be no degradation in performance due to respiration rate (mainstream)
AwRR accuracy	± 1 rpm
Operation mode	Measure, standby
Sample gas flow rate (sidestream)	(50 $\pm$ 10) ml/min

O₂ Compensation

Range	0% to 100%
Resolution	1%
Default	16%
Barometric pressure compensation	User setup

Anesthetic Gas Compensation

Range	0% to 20%
Resolution	0.1%
Default	0.0%
Balance gas compensation	Room air, N ₂ O, helium

Stability

Short-term drift	Drift over 4 hours < 0.8 mmHg
Long-term drift	120 hours
Zero calibration	Support
Alarm type	etCO ₂ , FiCO ₂ , AwRR
Apnea alarm delay	10 s, 15 s, 20 s, 25 s, 30 s, 35 s, 40 s; default value is 20 s
Data sample rate	100 Hz
CO ₂ rise time/response time (mainstream)	Less than 60 ms
Sensor response time (sidestream)	< 3 seconds, including transport time and rise time

Technical Data

Interfering Gas and Vapor Effects on etCO₂ Measurement Values:

Nitrous oxide	60	Dry and saturated gas
Halothane Enflurane	4	(0 ~ 40) mmHg: ± 1 mmHg additional error
Isoflurane	5	(41 ~ 70) mmHg: ± 2.5% additional error
Sevoflurane	5	(71 ~ 100) mmHg: ± 4% additional error
Xenon	5	(101 ~ 150) mmHg: ± 5% additional error
Helium	80	Note:
Desflurane	50	Additional worst case error when
	15	compensation for PB, O ₂ , N ₂ O, anesthetic
		agents, or helium is correctly selected for
		the actual fractional gas constituents
		present.
		Desflurane:
		The presence of desflurane in the exhaled
		breath at concentrations greater than 5%
		will positively bias carbon dioxide values
		by up to an additional 3 mmHg at 38
		mmHg.
		Xenon:
		The presence of xenon in the exhaled
		breath will negatively bias carbon dioxide
		values by up to an additional 5 mmHg at
		38 mmHg.

Barometric Pressure on etCO₂ Measurement Values:

Quantitative Effect

Ambient barometric, operational
(0 ~ 40) mmHg: ± 1 mmHg additional error
(41 ~ 70) mmHg: ± 2.5% additional error
(71 ~ 100) mmHg: ± 4% additional error
(101 ~ 150) mmHg: ± 5% additional error
Note: Additional worst case error when compensation for PB, O₂, N₂O, anesthetic agents, or helium is correctly selected for the actual fractional gas constituents present.

NOTE

Respiration rate accuracy was verified by using a solenoid test setup to deliver a square wave of known CO₂ concentration to the device. 5% and 10% CO₂ concentrations were used. Respiration rate was varied over the range of the device. Pass/fail criteria was comparison of the respiratory rate output from the sensor to the frequency of the square wave.

Dräger MCable Mainstream CO₂ Module

Measure parameters	etCO ₂ , FiCO ₂ , AwRR
Unit	mmHg, %, Kpa

Technical Data

Measuring Range

etCO ₂	0 mmHg to 100 mmHg	
FiCO ₂	0 mmHg to 100 mmHg	
AwRR	3 rpm to 150 rpm (PGM algorithm)	
Resolution	etCO ₂	1 mmHg
	FiCO ₂	1 mmHg
	AwRR	1 rpm
etCO ₂ accuracy	< 0.5 mmHg rms, 0 mmHg to 40 mmHg	
	< 1 mmHg rms, 40.1 mmHg to 100 mmHg	
Operation mode	Measure, standby	
Local barometric pressure	57 kPa to 110 kPa	

O₂ Compensation

Range	0% to 100%
Resolution	1%
Default	16%

N₂O Compensation

Range	0% to 100%
Resolution	1%
Default	0%

He Compensation

Range	0% to 100%
Resolution	1%
Default	0%

Xe Compensation

Range	0% to 100%
Resolution	1%
Default	0%
Zero calibration	Support
Alarm type	etCO ₂ , FiCO ₂ , AwRR
Apnea alarm delay	10 s, 15 s, 20 s, 25 s, 30 s, 35 s, 40 s; default value is 20 s
Data reporting rate	Every 10 msec or 20 msec

Technical Data

Response time	Rise time: t10-90 = 24 msec Delay time: 150 msec
Warm up	The sensor meets the specified operating performance within 2 minutes typical from power on or reset at ambient temperatures from 20°C to 40°C (68°F to 104°F). At 10°C (50°F) ambient temperature, time from power on to reach the specified operating performance is 10 min approximately.

Interfering Gases and Vapours

N ₂ O 100 vol.%	0.00 vol.%
Halothane 5 vol.%	0.02 vol.%
Enflurane 5 vol.%	0.03 vol.%
Isoflurane 5 vol.%	0.02 vol.%
Sevoflurane 5 vol.%	0.02 vol.%
Desflurane 20 vol.%	0.00 vol.%
Ethanol 4‰ *	0.00 vol.%
Acetone 1‰ *	0.00 vol.%
Isopropanol 1%	0.00 vol.%
Methane 3 vol.%	<0.02 vol.%
NO 100 ppm	0.01 vol.%
NO ₂ 50 ppm	0.00 vol.%
CO 4 vol.%	0.00 vol.%
Freon R21 100 vol.%	0.07 vol.%
Freon R134a 100 vol.%	0.19 vol.%
Heptafluoropropane 0.7 vol.%	0.00 vol.%
Water vapour 37°C saturated	0.01 vol.%
*blood concentration equivalent	

NOTE

The numbers given at the end of each line are typical CO₂ readings of the sensor for the pure interfering gas or vapour, balance N₂ (if applicable), without CO₂ content. CO₂ reading of common mixtures like CO₂, O₂, N₂O, anaesthetic agent (in physiological concentration) or CO₂, O₂, N₂, water vapour is within specified bias, provided that the major foreign gases (see above: O₂, N₂O, He, Xe) are entered to the sensor.

Effects of Humidity or Condensate

The airway adapter windows are indirectly heated via the sensor to prevent moisture condensation. While by sensor design the effect of water droplets spilled onto the airway adapter windows and of contamination, as long as still some measurement light passes the airway adapter windows, is largely compensated for, water droplets and other window contamination may slightly influence measurement bias, up to 0.3 Vol.% approximately at 5 Vol.% CO₂ (normally much less). Precision, of course, worsens if less light passes (i.e., noise of reading gets higher). After some time, water droplets are heated away. If measurement light is blocked such that noise of reading gets unacceptably high, an error message is sent from the CO₂ sensor indicating that the airway adapter has to be checked (cleaned or replaced).

Technical Data

BIS

Technique	Bispectral Index, Power Spectrum Analysis		
Measure parameters	Primary parameter	BIS	0 to 100
	Secondary parameters	SQI	0% to 100%
		SR	0% to 100%
		EMG	30 dB to 80 dB
		SEF	0.5 Hz to 30.0 Hz
		TP	40 dB to 100 dB
		BC (only applicable to BIS™ extend sensor)	0 to 30
Sweep speed	6.25 mm/s, 12.5 mm/s, 25 mm/s, 50 mm/s		
Wave scale	50 µV, 100 µV, 200 µV, 500 µV		
BIS trend	Length of BIS trend: 6 min, 12 min, 30 min, 60 min		
Smoothing rate	10 s, 15 s, 30 s		
Noise (EEG waveform)	< 0.3 µV (0.25 Hz ~ 50 Hz)		
EEG bandwidth	0.25 Hz ~ 50 Hz		
BIS alarm range	0 ~ 100		

C.O.

Measure parameters	C.O., TB, TI
Measurement method	Thermodilution technique

Measuring Range

C.O.	0.1 l/min ~ 20 l/min
TB	+23 °C to +43 °C (+73.4 °F to +109.4 °F)
TI	-1 °C to +27 °C (+30.2 °F to +80.6 °F)

Resolution

C.O.	0.1 L/min
TB, TI	+0.1 °C (+0.1 °F)

Accuracy

C.O.	±5% or ± 0.2 L/min, whichever is greater
TB	±0.1 °C (±32.18 °F) (without sensor)
TI	±0.1 °C (±32.18 °F) (without sensor)

Technical Data

Trend review

Short	1 hr, 1 s. resolution
Long	150 hrs, 1 min. resolution
NIBP measurement data review	1200 sets
Alarm review	200 sets
Arrhythmia review	200 sets

NOTE

Regarding the AG specifications, refer to the Supplement Scio Four modules.

Wireless

IEEE	802.11 b/g/n
Frequency band	2.4 GHz ISM band
Modulation	OFDM with BPSK, QPSK, 16-QAM, and 64-QAM 802.11 b with CCK and DSSS
Typical transmit power (± 2 dBm)	17 dBm for 802.11 b DSSS, 17 dBm for 802.11 b CCK, 15 dBm for 802.11 g/n OFDM

Device Connectivity

Protocol	Medibus/Medibus.X
Supported device	Supported ventilators: Evita V500, Evita V300, Savina 300, Savina, Babylog 8000, Babylog VN500, Oxylog 3000+, Evita V800, Evita V600, Babylog VN800, Babylog VN600; Supported anesthesia machines: Perseus A500, Zeus Infinity Empowered, Primus Infinity Empowered, Apollo, Fabius GS Premium, Fabius Tiro, Fabius MRI, Fabius plus, Fabius plus XL, Primus, Atlan.

Recorder

Record width	48 mm (1.9 inch)
Paper width	50 mm
Paper speed	12.5, 25, 50 mm/s
Trace	Up to 3 waveforms

Technical Data

Recording types	• Continuous real-time recording • 8/20 seconds real-time recording • Oxygenation calculation result recording • Ventilation calculation result recording • Renal function calculation result recording • Trend graph recording • Trend table recording • NIBP review recording • Arrhythmia review recording • Alarm review recording • C.O. measurement recording • Frozen waveform recording • Drug calculation titration recording • Hemodynamic calculation result recording
Display Specifications	
Display screen	380 mm (15 inch) color TFT
Resolution	1024 x 768
Maximum number of waveforms	13
Indicator LEDs	1 power, 2 alarm, 1 charge
Physical Specification	
Size (H x W x D)	(408±2) mm x (316±2) mm x (157±2) mm (12.4 x 16.1 x 6.2 inch)
Weight	<7.0 kg (15.4 lbs)
Electrical Specification	
Power supply	100 V – 240 V~, 50 Hz/60 Hz
Current	1.4 A-0.7 A
FUSE	T 3.15 AH, 250 VP
Classification	
Protection class	Class I equipment and internal powered equipment
EMC type	Class A
Degree of protection against electric shock	CF: ECG (RESP), TEMP, IBP, C.O. BF: SpO ₂ , NIBP, CO ₂ , AG, BIS
Liquid ingress protection	IPX1
Mode of operation	Continuous

Technical Data

Lithium-ion Battery (optional)

Quantity	1
Capacity	5,000 mAh
Battery life	≥ 300 min (At 20-30°C, with (a) new fully charged battery/ batteries, continuous SpO ₂ measurement and NIBP automatic measurement mode at interval of 15 minutes, Dräger ECG/TEMP module connected, recording at interval of 10 minutes, brightness set to "1")
Battery charge time	≤ 390 min, 100% charge (monitor is on or in standby mode) ≤ 351 min, 90% charge (monitor is on or in standby mode)

Environmental Requirements

The monitor may not meet the performance specifications given here if stored or used outside the specified temperature and humidity ranges. When the monitor and related products have differing environmental specifications, the effective range for the combined products is that range which is common to the specifications for all products.

Temperature Range

Operating	+15 °C to +27 °C (+59 °F to +80.6 °F)
Transport and storage	-10 °C to +40 °C (14 °F to +104 °F)

Relative Humidity

Operating	30%RH to 70%RH (non-condensing)
Transport and storage	5% RH to 70% RH (non-condensing)

Atmospheric Pressure

Operating	70 kPa to 101 kPa
Transport and storage	50 kPa to 105 kPa

Standards

IEC 60601-1: 2005+A1 :2012; IEC 60601-1-2: 2007; EN 60601-1: 2006+A1 :2013; EN 60601-1-2: 2007; IEC 60601-2-49: 2011 The Vista 120 monitors comply with the Medical Device Directive (MDD) 93/42/EEC.

Vista 120	MS34008	MS34010	MS34009	MS34011
3/5 lead ECG	X	X	X	X
Proprietary SpO ₂	X		X	
Nellcor SpO ₂		X		X
NBP	X	X	X	X

Technical Data

Dual temps	X	X	X	X
3IBP			X	X
CO			X	X
etCO ₂			X	X
BISx			X	X
Built-in recorder		X	X	X
Gas bench	X	X	X	X
LAN	X	X	X	X
Wireless	X	X	X	X

Vista 120 monitors are available in select markets only.

For availability in your area, please contact the appropriate Dräger office from those listed below.

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